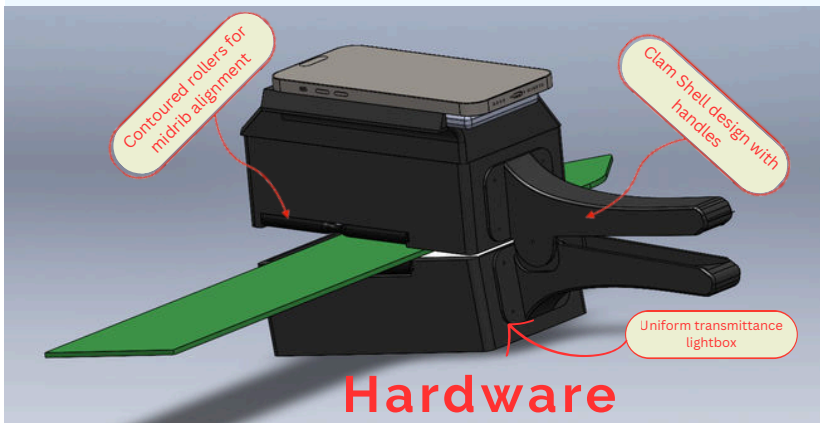


CHARLES WANG

BIOLOGICAL ENGINEERING AT PURDUE UNIVERSITY

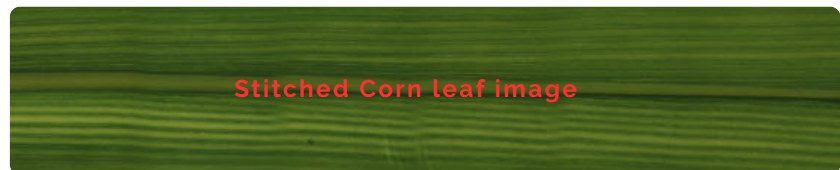
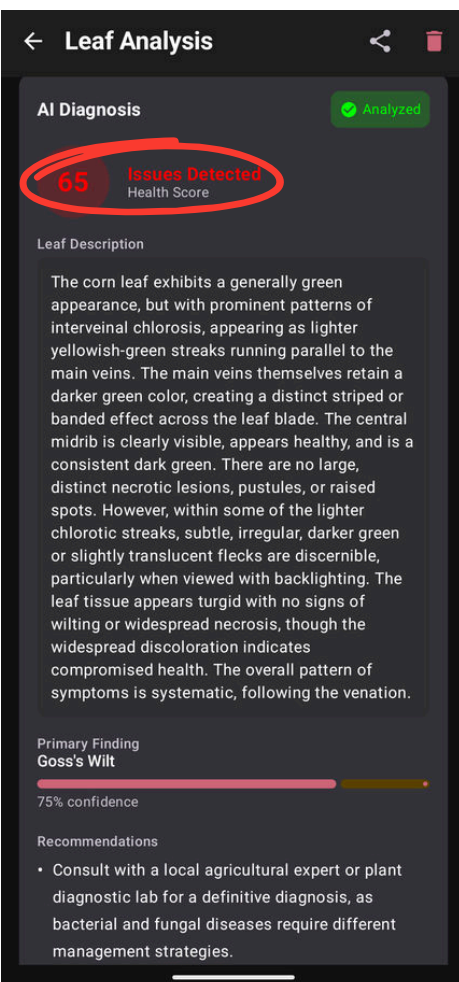
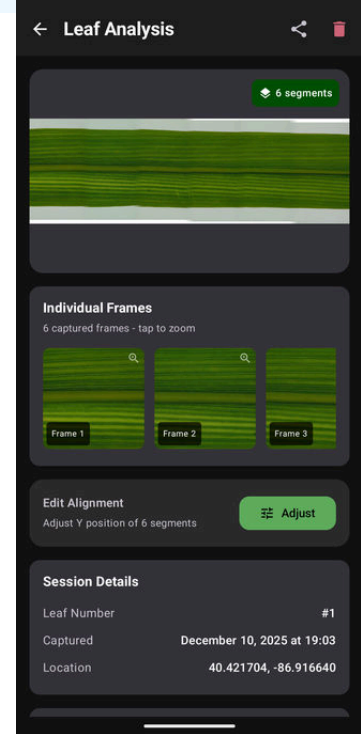
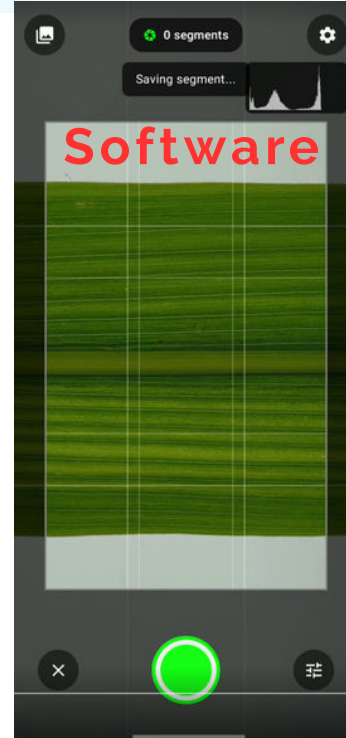
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SMARTPHONE PLANT IMAGING SYSTEM WITH AI-POWERED DIAGNOSIS - LEAFDOC



What?

- **\$50** handheld lightbox smartphone attachment that blocks ambient light and provides uniform LED illumination
- Contoured rollers **flatten** and center the sample on entry, ensuring repeatable positioning without manual adjustment
- Lightbox blocks external noise and provides **uniform** illumination
- **MagSafe** mounting locks smartphone above the leaf in a fixed, repeatable position



How?

- Modeled and iterated through 7 enclosure revisions in **SolidWorks**, 3D printing each to test light leak, roller clearance, and component fit
- Wrote an alignment stitching algorithm that assembles overlapping frames into a single full-length image
- Built a native **Android app** (Kotlin, CameraX) with manual camera controls and a multi-provider AI diagnostic pipeline

Results

- User captures image using smartphone then can organize and manage on app
- Cloud based AI delivers a **health score**, primary finding, and confidence level in-app, no export or external tools needed

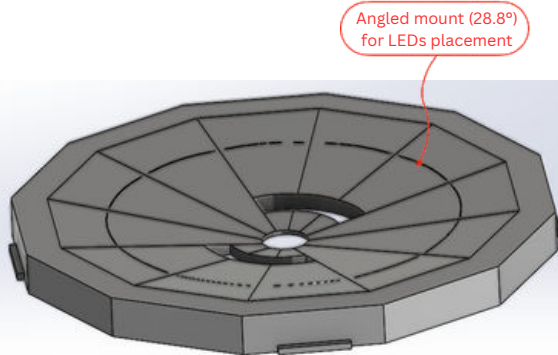
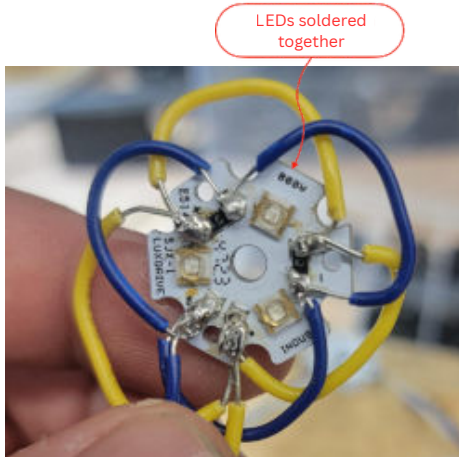
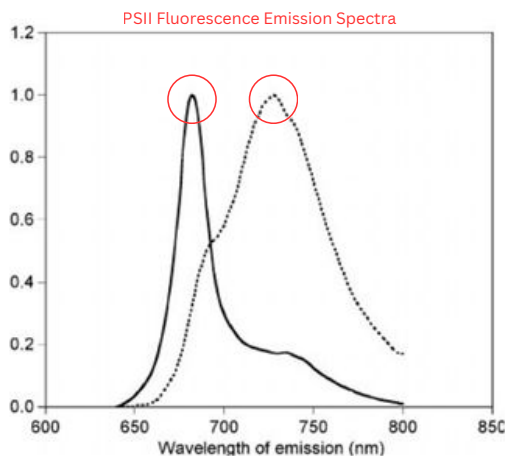


PORTABLE PHOTOSYNTHETIC EFFICIENCY IMAGING SYSTEM - FLUROSCAN



What?

- Direct measurement of photosynthetic efficiency
- Real-time non-invasive imaging with **early detection** of plant stress
- More affordable and accessible technology for the masses

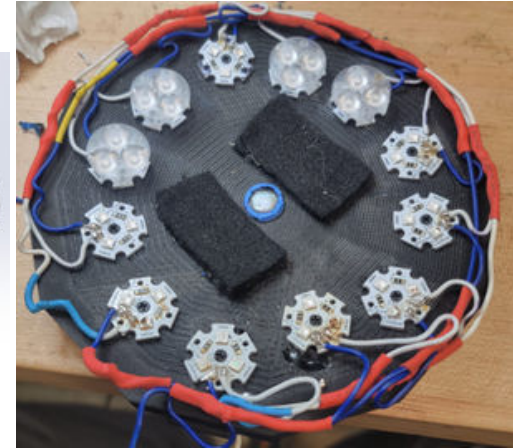


How?

- Used **Solidworks** to prototype
- Used **Raspberry Pi** for camera and system control
- Used **Bambu Labs** to 3D print
- Dark adapted leaves to fully open PSII
- Saturated photosystems with 450 nm LEDs and captured images through a 680/720 nm filter

Theory

- A weak, non-actinic light will excite and open PSII reaction centers and give the **baseline fluorescence** (F_o)
- A strong pulse of light will close all PSII reaction centers and give **maximum fluorescence** (F_m)
- **Ratio** represents maximum efficiency of PSII energy conversion



Specifications

- Achieved 450 nm excitation light at 4000 $\mu\text{mol m}^{-2}\text{s}^{-1}$ using 12 LEDs in 3-up configuration

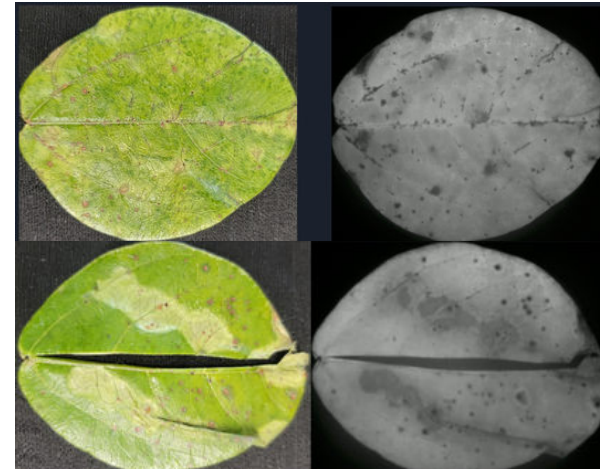
$$F_v = F_m - F_o$$

$$F_v / F_m$$

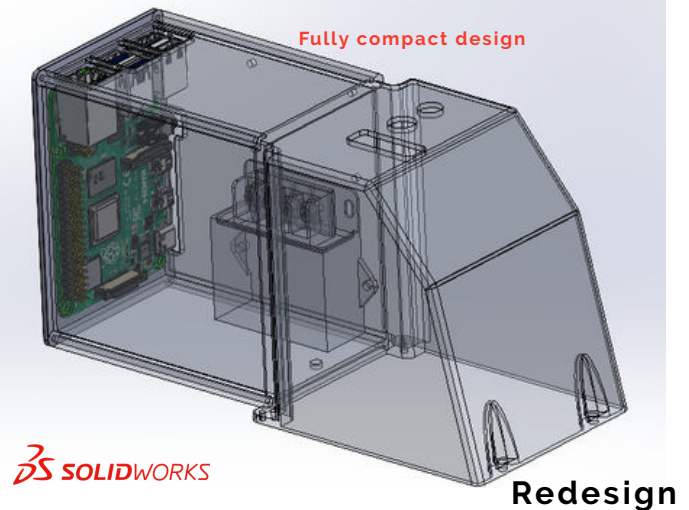
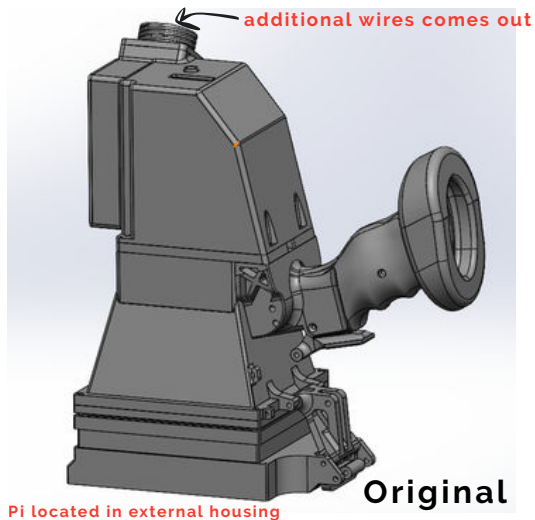


Control

Results



MULTISPECTRAL HANDHELD IMAGING DEVICE



 SOLIDWORKS

What?

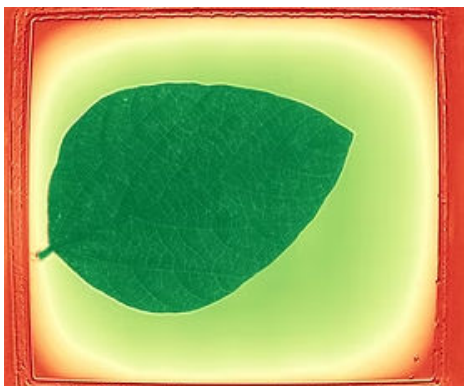
- Plant phenotyping device that collects **high resolution images** (0.04mm/pixel)
- Redesigned enclosure houses all electronics internally, eliminating external wiring for a fully field-portable system

How?

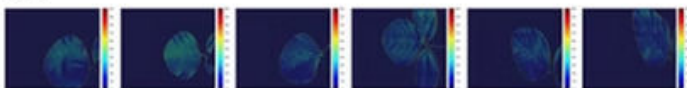
- Used **Solidworks** to prototype
- Used **Bambu Labs** to 3D print
- Used **Raspberry Pi** for camera and system control
- Reverse-engineered and documented all subsystems
- Soldered 120+ LEDs across 6 wavelength bands with MOSFET transistors and 12V/5V power conversion

Results

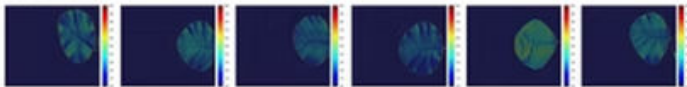
- High signal to noise ratio** with ambient light blocked
- Clearly captured the secondary veins on leaves
- Detected plant stress days before visual symptoms
- Enabled co-researcher's ML model to achieve 90% classification accuracy in a peer-reviewed publication (co-author)



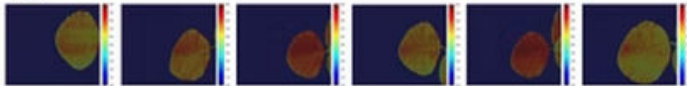
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INSTITUTE FOR DIGITAL FORESTRY

LeafSpec® Multispectral Handheld Imaging Device

Charles Wang And Aiden Mo



MEASURING EVERY TREE ON THE PLANET

ABSTRACT

Multispectral imaging (MSI) technology has been widely applied in plant phenotyping due to its capability to detect various plant properties through specific spectral bands. This study presents the development of a portable multispectral handheld device designed for leaves imaging. The device, characterized by its ease of use, isolates the leaf eliminating outside factors; this provides a high signal-to-noise ratio. This device can capture details down to 0.04 millimeters allowing for analysis at tissue level. This information can be leveraged to parse spatial and spectral features from the images. We utilize these additional features to find new plant stress signals. This enables the use of AI to develop early disease detection models.

PRESENTER BIO INFORMATION

Charles Wang
Undergraduate Biological Engineering



Aiden Mo
Undergraduate DATA (Data Analysis, Technologies, and Applications)

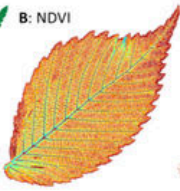


Scan	Process	Results
Portability - Lightweight and Portable - Handheld - Non-destructive - Affordable - Quick swap battery pack	Artificial Intelligence - Automatic extraction of spatial and spectral features - Automatic detection of diseases - Monitoring Feature Importance	Helps Environment - Less yield loss - Detects ideal fertilizer usage (20%) - Reduces runoff
Convenience - One click - Fast scanning (Multispectral in seconds) - Wireless uploading - Automatic GPS tracking	Higher Quality Data - High signal-to-noise ratio - Higher quality images - Complete isolation from environment - Uniform light source	Cloud Based - Cloud based storage - All data accessible - Central collection source - Leverages global data

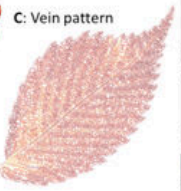
A: RGB



B: NDVI



C: Vein pattern







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